# SEQUENCES & SERIES

Arithmetic &
Geometric
Progressions

# What Are Sequences & Series?

Let's Learn Together!

A sequence is a list of numbers arranged in a special order, following a pattern or rule.

A series is what we get when we add up the terms of a sequence.

Example: 2, 4, 6, 8... is a sequence

2 + 4 + 6 + 8 = 20 is a series!

# Understanding Sequences

2, 4, 6, 8, ...

# What is an Arithmetic Progression (AP)?

Numbers increase or decrease by the same amount each time.

Formula: $a_n = a_1 + (n-1)d$

- $a_1$ = first term
- $d$ = common difference
- $n$ = term number

# Example of AP

Sequence: 3, 6, 9, 12, ...

$$a_5 = a_1 + (n-1)d$$

$$a_5 = 3 + (5-1) \times 3$$

$$a_5 = 3 + 12 = 15$$

# Sum of an AP Series

$$S_n = n/2 \times (2a_1 + (n-1)d)$$

# What is a Geometric Progression (GP)?

"Numbers multiply or divide by the same number each time."

Formula: $a_n = a_1 \times r^{(n-1)}$

- $a_1$ = first term
- $r$ = common ratio
- $n$ = term number

## Example of GP

2, 6, 18, 54, ...

# Sum of a GP Series

Formula: $S_n = a_1 \times (r^n - 1)/(r - 1), r \neq 1$

Example: Sum of first 3 terms of 2, 6, 18

$S_3 = 2 \times (3^3 - 1)/(3 - 1) = 2 \times 26/2 = 26$

# AP vs GP: What's the Difference?

Arithmetic Progression (AP):

- Add or subtract the same number
- Linear growth: 2, 5, 8, 11...
- Formula: $a_n = a_1 + (n-1)d$

Geometric Progression (GP):

- Multiply or divide by the same number
- Exponential growth: 2, 6, 18, 54...
- Formula: $a_n = a_1 \times r^{(n-1)}$

## Let's Compare!

# Practice Problems: AP

1. Find the 7th term of 5, 8, 11, ...

2. Find the sum of first 10 terms of 2, 4, 6, ...

3. Find the common difference in 15, 12, 9, ...

# Practice Problems: GP

1. Find the 5th term of 3, 9, 27, ...

2. Find the sum of first 4 terms of 1, 2, 4, 8, ...

3. Identify the common ratio of 81, 27, 9, ...

# Solutions: AP Practice

7th term of 5, 8, 11...

$a_7 = 5 + (7-1) \times 3 = 23$

Sum of first 10 terms of 2, 4, 6...

$S_{10} = 10/2 \times (2 \times 2 + 9 \times 2) = 110$

Common difference of 15, 12, 9...

$d = 12 - 15 = -3$

# Solutions: GP Practice

5th term of 3, 9, 27...

$a_5 = 3 \times 3^4 = 243$

Sum of 1, 2, 4, 8...

$S_4 = 1 \times (2^4-1)/(2-1) = 15$

Ratio of 81, 27, 9...

$r = 27 \div 81 = 1/3$

# Let's Review! Quiz Time

In AP 4, 7, 10, ... the common difference is ___

The 3rd term of GP 2, 6, 18 is ___

AP formula: $a_n = a_1 + (n-1) \times$ ___

# Thank You for Learning!